

The Burrau–Pythagorean Family

A Research Programme for Structured Exploration
of the Three-Body IC Manifold

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Abstract

We propose a structured exploration of the three-body problem’s initial-condition (IC) space via the *Burrau–Pythagorean family*: the infinite set of right-triangle configurations whose side lengths form Pythagorean triples, with masses assigned by the classical Burrau convention. This family is cleanly enumerable via Euclid’s parametrisation, admits a continuous interpolation controlled by a single ratio $\nu = n/m \in (0, 1)$, and maps directly into the Principia IC pipeline. The document outlines the mathematical construction, a progressive escalation from discrete survey through continuous geometry and mass/momentum relaxation to the full 10D manifold, formal chart maps for each stage, a testable central hypothesis connecting number theory to Hamiltonian dynamics, and connections to the symbolic dynamics and ML research programme. Three papers on arXiv mention Burrau’s problem; none survey the Pythagorean family systematically.

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Part I

The Family

0.1 Motivation

Burrau’s problem (1913): three bodies at the vertices of a 3–4–5 right triangle, each mass equal to its opposite side. It is one of the most studied chaotic trajectories in celestial mechanics, and its fractal basin boundaries are a canonical example of sensitive dependence on initial conditions.

But 3–4–5 is the simplest member of an infinite family. Every primitive Pythagorean triple (a, b, c) defines a distinct right-triangle Burrau configuration. The Burrau mass convention $m_i = \text{opposite side}$ (normalised to $\sum m_i = 1$) extends naturally to all of them.

The gap. Three papers on arXiv mention Burrau’s problem (Chitan & Mylläri 2020, 2024; Lousto & Zlochower 2007). All study relativistic extensions of the single 3–4–5 case. No one has surveyed the Newtonian Pythagorean family systematically. No one has asked whether the 3–4–5 case is dynamically special among all right-triangle Burrau configurations, or whether the fractal basin structures persist when the Pythagorean constraint is relaxed.

0.2 Euclid’s parametrisation

For coprime integers $m > n > 0$ with $m - n$ odd:

$$\boxed{a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2.}$$

Every primitive Pythagorean triple arises exactly once (up to leg swap). Non-primitive triples (ka, kb, kc) are dynamically equivalent under the similarity symmetry of Newtonian gravity ($\mathbf{r} \rightarrow \lambda \mathbf{r}$, $t \rightarrow \lambda^{3/2} t$).

Distinct shapes. Different primitive triples define genuinely non-similar triangles:

(m, n)	Triple (a, b, c)	Acute angle	Aspect a/b
(2, 1)	(3, 4, 5)	36.9°	0.75
(3, 2)	(5, 12, 13)	22.6°	0.42
(4, 1)	(15, 8, 17)	61.9°	1.88
(4, 3)	(7, 24, 25)	16.3°	0.29
(5, 2)	(21, 20, 29)	46.4°	1.05
(5, 4)	(9, 40, 41)	12.7°	0.23

Each represents a qualitatively distinct initial configuration with different mass ratios, body separations, and potential energy.

Enumeration. The number of primitive triples with $c \leq N$ grows as $\mathcal{O}(N/\log N)$. For $m \leq 50$: approximately 400 primitive triples, more than sufficient for a systematic survey.

0.3 Normalised Burrau configuration

Fix the right angle at the origin, normalise by the hypotenuse:

$$\mathbf{r}_1 = (0, 0), \quad \mathbf{r}_2 = \left(\frac{a}{c}, 0\right), \quad \mathbf{r}_3 = \left(0, \frac{b}{c}\right).$$

Burrau masses:

$$(m_1, m_2, m_3) = \frac{1}{a+b+c}(c, b, a).$$

Jacobi vectors (inner pair 1–2):

$$\boldsymbol{\rho} = \left(\frac{a}{c}, 0\right), \quad \boldsymbol{\lambda} = \left(-\frac{ab}{c(b+c)}, \frac{b}{c}\right).$$

These feed directly into the Principia decoder/canonicaliser with no special-case logic.

0.4 Continuous interpolation

Nothing in Euclid’s formulae requires m, n to be integers. For any real $m > n > 0$, $a^2 + b^2 = c^2$ holds identically. The single ratio

$$\nu := \frac{n}{m} \in (0, 1)$$

determines the triangle shape completely (normalisation by c removes overall scale). Normalised Euclid formulae:

$$a(\nu) = 1 - \nu^2, \quad b(\nu) = 2\nu, \quad c(\nu) = 1 + \nu^2.$$

The acute angle sweeps monotonically:

$$\theta(\nu) = \arctan\left(\frac{1 - \nu^2}{2\nu}\right) = \arctan\left(\frac{a}{b}\right).$$

Inversion: $\nu(\theta) = \sec \theta - \tan \theta$.

All positions, Jacobi vectors, and Burrau masses vary smoothly with ν . Primitive Pythagorean triples are countable landmarks on a smooth 1D landscape. Whether these landmarks are dynamically special—local extrema of chaos measures, bifurcation points in basin topology—or unremarkable members of a smooth family is the first scientific question.

Part II

Progressive Escalation

The degrees of freedom decompose cleanly into geometry, mass, and momentum. Each can be relaxed independently, producing a natural escalation from 0D to 10D.

0.5 Stage 1a — Discrete survey

What. The strict Burrau–Pythagorean family at rest. For each primitive (m, n) with $m \leq 50$, render the fractal basin map at zero momentum.

Scientific questions.

- How does basin topology (escape prevalence, collision structure, bounded-orbit regions) vary across primitive triples?
- Is there a systematic dependence on the aspect ratio a/b or the acute angle θ ?
- Do any triples exhibit qualitatively different behaviour (e.g. a triple where body 3 never escapes, or where bounded orbits dominate)?

Output. A catalogue of basin maps indexed by (m, n) . Already novel: no systematic survey exists.

0.6 Stage 1b — Continuous geometry

What. Sweep $\nu \in (0, 1)$ continuously at zero momentum, Burrau masses.

Chart map.

$$\Phi_{\text{Euclid}}(u, v) = (\nu(v), \mathbf{m}_{\text{Burrau}}(\nu(v)), \text{rest}),$$

with $\nu(v) = \nu_{\min} + (\nu_{\max} - \nu_{\min})v$ and the horizontal axis displaying m as a logarithmic landmark coordinate.

Scientific questions.

- Do primitive Pythagorean triples sit at dynamically special values of ν ?
- Is there continuous variation in basin structure, or are there bifurcations at specific ν values?
- Does the basin topology change character at extreme aspect ratios ($\nu \rightarrow 0$ or $\nu \rightarrow 1$)?

Output. A bifurcation diagram: basin statistics (escape fractions, collision fractions, bounded fractions, mean event time, mean word length) as functions of ν , with primitive triples highlighted.

0.7 Stage 2a — Mass relaxation

What. Fix a geometry via ν_0 , sweep the mass simplex Δ_2 at zero momentum.

Chart map.

$$\Phi_{\text{mass}}(u, v) = (\nu_0, (m_1, m_2, m_3), \text{rest}),$$

with $(u, v) \rightarrow \Delta_{2,\varepsilon}$ via the standard triangle warp plus interior buffer.

Scientific questions.

- How do basins deform as masses are dragged away from the Burrau assignment?
- Is the Burrau point dynamically special within the simplex (e.g. a critical point of some chaos measure, a local extremum of escape prevalence)?
- Equal masses $(1/3, 1/3, 1/3)$ sit at the barycentre. How does the basin structure at the barycentre compare with the Burrau point for each triangle shape?

Output. Ternary basin maps for selected ν_0 values, with the Burrau point and equal-mass point marked. Animatable across ν .

0.8 Stage 2b — Momentum exploration

What. Fix geometry and Burrau masses, sweep momentum via (L_z, K) or (L_z, E) .

Chart map.

$$\Phi_{\theta, K}(u, v) = (\nu(\theta(u)), \mathbf{m}_{\text{Burrau}}(\nu(\theta(u))), (L_z=0, K(v))),$$

with $K(v) = K_{\max} v^{\gamma_K}$ and variants for (L_z, E) , fixed- K with swept L_z , or free 4D Jacobi momenta.

Scientific questions.

- The classical Burrau problem sits at $(L_z, K) = (0, 0)$: the corner of the feasibility wedge. How do basins nucleate and grow as energy and angular momentum increase?
- At what energy does the fractal structure simplify (basins become large and smooth)?
- Is there a critical angular momentum above which the Burrau-type basin structure disappears?

Output. (L_z, K) basin maps for selected triples. Bifurcation strips: θ on horizontal, K on vertical.

0.9 Stage 3 — Combined

What. Masses and momenta vary simultaneously. The Burrau rest-start point is a single landmark in a space of dimension up to $2 + 4 = 6$.

Scientific questions.

- Are there correlations between mass assignment and momentum structure invisible when each is varied in isolation?
- Does the Burrau mass convention produce qualitatively different basin structure from equal masses at the same energy?

0.10 Stage 4 — Full 10D

What. Remove all constraints. The Burrau family becomes particular cross-sections of the full symmetry-reduced 10D IC manifold.

Scientific value. The Burrau family provides interpretable reference points within the high-dimensional space. Any structure discovered in the Burrau slices can be tracked into the unconstrained manifold via the projective microscope.

Part III

The Central Hypothesis

0.11 Statement

Hypothesis. Fractal basin structures observed in the Burrau–Pythagorean family are cross-sections of codimension- k structures ($k < 8$) in the full 10D IC manifold, rather than artefacts of the constrained submanifold. Basin boundaries that appear as 1D curves in the 2D Burrau slice persist as higher-dimensional surfaces when the slicing plane is rotated into unconstrained directions.

0.12 How to test it

The projective microscope provides a direct test. Lock onto a basin boundary identified in a Burrau slice—a point $\mathbf{z}_0$ on or near a boundary between, say, escape-body-1 and escape-body-2 basins. Then:

1. Begin with the slice basis spanning the Burrau-constrained submanifold.
2. Continuously rotate one basis vector into an unconstrained direction (e.g. independent mass variation, or momentum perturbation) using the tilt mechanism.
3. Observe whether the basin boundary persists, deforms smoothly, or dissolves.

Positive result (persistence). The boundary deforms smoothly as the slice rotates. This establishes that number-theoretic initial conditions—the Pythagorean triples—are identifying dynamically meaningful structures in the full IC space. A connection between number theory and Hamiltonian dynamics that has not been previously demonstrated.

Negative result (dissolution). The boundary dissolves when the slice leaves the constrained submanifold. This characterises the Burrau family as a dynamically special (measure-zero) subset with qualitatively different basin topology from generic nearby configurations. Also informative: it means the classical Burrau problem is more “fragile” than it appears.

Mixed result (partial persistence). Some boundaries persist, others dissolve. The persistent ones are genuine codimension- k structures; the dissolving ones are artefacts. Classifying which is which—and correlating with the free-group word of the boundary trajectories—is a research programme in itself.

0.13 The free-group word as a boundary diagnostic

The free-group word (defined in the implementation spec) provides a sharper tool for testing the hypothesis than metric-based spread alone.

Topological basin boundaries. At a metric basin boundary, nearby ICs produce different outcome classes. At a *topological* basin boundary, nearby ICs produce different free-group words. The topological boundary is generically finer—two trajectories can share an outcome class but differ topologically.

Three-level persistence diagnostic. When rotating the slice:

- If the *word* boundary persists: the topological structure is genuine.
- If only the *outcome* boundary persists but the word boundary dissolves: the metric boundary is robust but the fine topological structure is a slicing artefact.

- If both dissolve: the entire structure is constrained-submanifold-specific.

This diagnostic is only possible because the free-group word is computed and stored per sample.

Part IV

Connection to the ML Programme

The Burrau family is the natural first dataset for the Principia ML research programme (see companion document).

0.14 Why this family first

- **Structured.** The single parameter ν provides a clean axis along which to study how ML model accuracy varies with dynamical complexity.
- **Well-referenced.** The 3–4–5 case is the most studied three-body configuration; results can be validated against existing literature.
- **Progressive.** Stages 1a–4 provide datasets of increasing dimensionality, naturally testing how models scale with IC-space complexity.
- **Hypothesis-connected.** The central hypothesis is directly testable by ML models trained on Burrau data and evaluated on off-family data (transfer learning as a persistence test).

0.15 ML applications per stage

Stage	Dataset	ML question
1a	Discrete survey	Does word $\rightarrow$ outcome accuracy vary across triples?
1b	Continuous ν	How does the predictive horizon $\alpha(k)$ depend on ν ?
2a	Mass simplex	Does the model generalise from Burrau masses to arbitrary masses?
2b	Momentum	Does adding kinetic energy change which encounters are decisive?
3	Combined	Can a single model handle joint mass-momentum variation?
4	Full 10D	Does a Burrau-trained model transfer to the full manifold?

The Stage 4 transfer question is the ML analogue of the central hypothesis: if a model trained on Burrau family data performs well on off-family ICs, the dynamical structures it learned are genuine features of the full manifold.

Part V

Paper Plan

0.16 Two-paper strategy

Paper A — The Burrau–Pythagorean family (implementation + survey).

1. Introduction: Burrau’s problem, the Pythagorean family, the gap in the literature.
2. Mathematical construction: Euclid parametrisation, continuous interpolation, chart maps.
3. The Principia tool: brief description of the WebGPU manifold explorer.
4. Stage 1a results: discrete survey of basin maps across primitive triples.
5. Stage 1b results: continuous ν sweep, bifurcation diagram, are Pythagorean triples special?
6. Stage 2a: mass relaxation, ternary maps, Burrau vs equal-mass comparison.
7. Stage 2b: momentum exploration, (L_z, K) maps, basin nucleation.
8. The central hypothesis: projective microscope test, persistence analysis.
9. Discussion and conclusions.

Paper B — Symbolic dynamics and ML (the free-group word).

1. Introduction: symbolic dynamics, the free group F_2 , the encounter word.
2. Dataset: Principia-generated (word, outcome) corpus from the Burrau family.
3. Stage 0: classical feature baselines, feature importance.
4. Stage 1: word $\rightarrow$ outcome transformer, accuracy, attention analysis.
5. Stage 2: predictive horizon curve $\alpha(k)$.
6. Transfer: Burrau-trained model evaluated on off-family ICs.
7. Discussion: what encounter topology tells us about three-body dynamics.

Paper A establishes the family and the tool. Paper B uses the tool’s data for ML. They can be written in parallel; Paper A should appear first.

0.17 Key figures for Paper A

1. Grid of basin maps for the first 20 primitive triples, coloured by outcome.
2. Continuous bifurcation diagram: basin fractions vs ν , with primitive triples highlighted.
3. Ternary mass maps for $(3, 4, 5)$, $(5, 12, 13)$, $(7, 24, 25)$ with Burrau and equal-mass points marked.
4. (L_z, K) basin map for the 3–4–5 case, showing basin nucleation from rest.
5. Projective microscope sequence: a basin boundary persisting (or dissolving) as the slice rotates into unconstrained dimensions.
6. Free-group word length map for the 3–4–5 case, showing topological complexity structure.

0.18 Timeline

Phase	Activity	Duration
1	Build Principia v1 (from implementation spec)	4–8 weeks
2	Stage 1a discrete survey	1 week
3	Stage 1b continuous sweep + bifurcation diagram	1 week
4	Stage 2a/2b mass and momentum exploration	2 weeks
5	Hypothesis test via projective microscope	1–2 weeks
6	Write Paper A	2–3 weeks
7	ML experiments (Stages 0–2)	2–3 weeks
8	Write Paper B	2–3 weeks

Phase 1 is the bottleneck. Phases 2–5 are fast because the tool does the work. Phases 6–8 can overlap.